

ACTIVITY3

Facing and Center Drilling

Names: \_\_\_\_\_  
\_\_\_\_\_  
(Last Name, First Name MI.)

Date: \_\_\_\_\_  
Start Time: \_\_\_\_\_  
End Time: \_\_\_\_\_

Instructor: Nico O. Aspra

*Note: When printing the worksheet, print the Data Collection up to the Analysis section back-to-back on a single sheet of paper. Print the Rating Sheet on a separate sheet and staple it at the back of the worksheet.*

Table 3.1: Operational Parameters and Surface Finish Results

| Spindle Speed | Depth of Cut | Feed Rate | Finish |
|---------------|--------------|-----------|--------|
|               |              |           |        |
|               |              |           |        |
|               |              |           |        |
|               |              |           |        |
|               |              |           |        |
|               |              |           |        |

\* The finish should be rated as rough, grooved, matte, glossy, or mirror-like depending on the observed surface quality.

Table 3.2: Comparison of Machined Dimensions with Design Specifications

| Dimension         | Specification | Actual | Deviation | Remarks |
|-------------------|---------------|--------|-----------|---------|
| length            |               |        |           |         |
| center hole depth |               |        |           |         |

\* The “Remarks” column will be filled in by your instructor based on inspection.

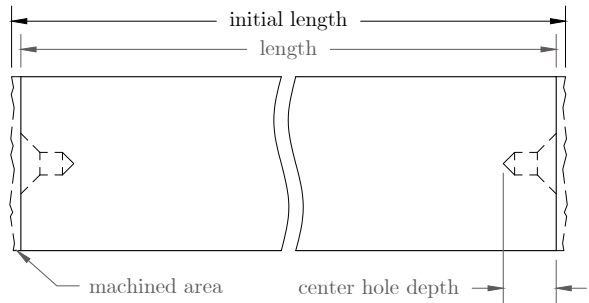


Figure 3.1: Final product plan used as a reference for dimensional measurements recorded in Table 3.2.

3.1

Analysis and Discussion

Reflect on the exercise and draw upon both your experience and the data gathered to respond to the following questions. Support your answers with specific examples from your observations.

Solutions

Why is it essential to manually rotate the chuck one full revolution before turning on the lathe? What issues might this step help prevent?

Question 2

Which combination of spindle speed, depth of cut, and feed rate gave the best surface finish during facing? Why do you think this setting worked well?

Question 3

In your experience, what factor had the most noticeable effect on surface finish—speed, feed, tool sharpness, or setup? How can this be optimized?

Question 4

What is the purpose of creating a pilot hole before center drilling? How does it improve drilling accuracy?

Question 5

How did you measure and mark the workpiece length before cutting? Were the results accurate? If not, what could have caused the deviation?

Contribution Rating Sheet

In this section, list and briefly describe each member’s contributions to the activity. Itemize the specific tasks performed and assign a corresponding percentage to each member. The combined percentage must total 100%.

| Name  | Designation<br>(Leader/Member) | Individual<br>Accomplishments | %    | Signature |
|-------|--------------------------------|-------------------------------|------|-----------|
|       |                                |                               |      |           |
|       |                                |                               |      |           |
| Total |                                |                               | 100% |           |

Academic Honesty Statement

I/We hereby certify that I/we have written and developed this report. I/We affirm that the report I/we am/are submitting as part of the requirements of this course is original and not plagiarized. My/Our signature/s below constitute/s my/our pledge that I/we have fully complied with Bicol University’s policy on academic integrity. I/We understand that academic dishonesty will not be tolerated and that, if such instance/s are found and proven in this submitted work, a final grade of 5.0 will automatically be given to me/us, and I/we will be subjected to disciplinary action/s sanctioned by Bicol University.

Signature over printed name (Group Leader)

Do not write beyond this point. This section will be completed by the instructor.

| Criteria                | 4 – Exemplary                                                                                | 3 – Proficient                                                        | 2 – Developing                                                                | 1 – Beginning                                                | Score |
|-------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------|-------|
| Understanding of Task   | Demonstrates complete understanding of the objectives, theory, and relevance of the activity | Shows good grasp of the task with minor conceptual gaps               | Basic understanding with some confusion about the purpose or process          | Limited or incorrect understanding of the task’s goal        |       |
| Execution Accuracy      | All procedures and tools are correctly used with high precision and consistency              | Most steps are followed correctly with minor errors or inefficiencies | Several key steps missed or tools used with noticeable inaccuracy             | Process poorly executed; improper use of tools or procedures |       |
| Measurements            | Measurements are accurate, clearly recorded, and well-analyzed against design targets        | Mostly accurate data with partial analysis or incomplete comparison   | Data is somewhat inaccurate or poorly explained                               | Lacks measurements or data is irrelevant or incorrect        |       |
| Reflection and Analysis | Deep insights, thoughtful evaluation of outcomes, and strong suggestions for improvement     | Reflection shows good understanding with reasonable suggestions       | Limited self-assessment or vague comments                                     | Little to no reflection; fails to engage with outcomes       |       |
| Presentation            | Report is highly organized, clear, and free of major errors in structure or expression       | Report is generally clear and well-organized with minor lapses        | Report lacks clarity or organization; some confusion in formatting or writing | Disorganized or incomplete submission; difficult to follow   |       |
| Total                   |                                                                                              |                                                                       |                                                                               |                                                              |       |